PARISHRAM 2.0 2026 (Class 12th)

Physics

DPP:01

Electric Charges and Fields

Q1 The unit of electric field intensity is.....
(A) N/C
(B) V/m
(C) Both (A) and (B)
(D) None of these

Q2 The field due to a charge at a distance 'x' from it is 'E'. When the distance is doubled, the intensity of the field is
(A) E/8 (B) E
(C) 2E (D) E/4

Q3 The magnitude of electric field intensity *E* is such that, an electron placed in it would experience an electrical force equal to its weight is given by
(A) mge (B) $\frac{mg}{e}$
(C) $\frac{e}{mg}$ (D) $\frac{e^2}{m^2}g$

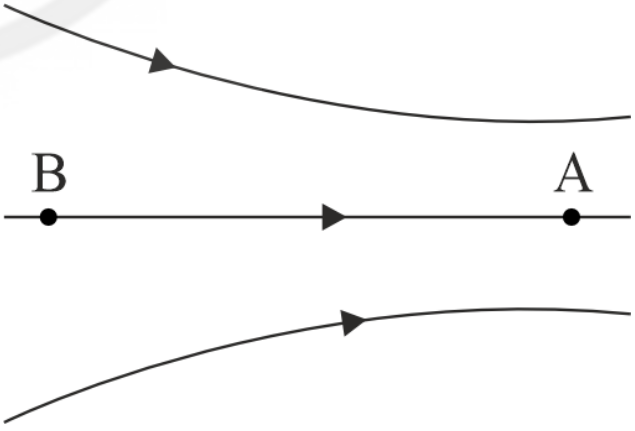
Q4 A drop of $10^{-6}\ kg$ water carries $10^{-6}C$ charge. What electric field should be applied to balance its weight (assume $g = 10\ m/s^2$)
(A) 10 V/m upward
(B) 10 V/m downward
(C) 0.1 V/m downward
(D) 0.1 V/m upward

Q5 Which of the following statements about electric lines of force is false?
(A) They always start at positive charges and end at negative charges.
(B) They form closed loops.
(C) They represent the path a positive test charge would follow in an electric field.
(D) The density of lines of force increases with the strength of the electric field.

Q6 Which of the following is true about the electric lines of force?
(A) They never intersect each other.
(B) They form closed loops.
(C) They always point towards the positive charge.
(D) They do not depend on the magnitude of the charge.

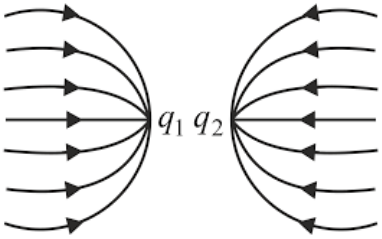
Q7 The electric field lines due to a positive point charge:
(A) Point radially outward from the charge.
(B) Point radially inward towards the charge.
(C) Form closed loops around the charge.
(D) Are parallel to each other.

Q8 Some electric lines of force are shown in figure, for point A and B.



(A) $\mathbf{E_A} > \mathbf{E_B}$
(B) $\mathbf{E_A} < \mathbf{E_B}$
(C) $\mathbf{E_A} = \mathbf{E_B}$
(D) Can't determine from the given information

Q9 The given figure gives electric lines of force due to two charge q_1 and q_2. What are the signs of the two charges?



(A) Both are negative
(B) Both are positive
(C) q_1 is positive but q_2 is negative
(D) q_1 is negative but q_2 is positive

Q10 **Assertion:** The electric field lines between two point charges, one positive and one negative, are straight lines that form a direct path from the positive charge to the negative charge.
Reason: The electric field lines are always directed from positive to negative charges, and their symmetry depends on the relative magnitude and distance between the charges.
(A) Both assertion and reason are true, and the reason is the correct explanation of the assertion.
(B) Both assertion and reason are true, but the reason is not the correct explanation of the assertion
(C) Assertion is true, but reason is false.
(D) Assertion is false, but reason is true.

Answer Key

Q1	C
Q2	D
Q3	B
Q4	A
Q5	B
Q6	A
Q7	A
Q8	A
Q9	A
Q10	D

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 **Text Solution:**

Both (A) and (B)

Video Solution:

Q2 **Text Solution:**

The intensity of the field when the distance is doubled is $\frac{E}{4}$.

Video Solution:

Q3 **Text Solution:**

According to the question,

$eE = mg \Rightarrow E = \frac{mg}{e}$

Video Solution:

Q4 **Text Solution:**

By using $QE = mg$

$\Rightarrow E = \frac{mg}{Q} = \frac{10^{-6} \times 10}{10^{-6}} = 10\ V/m$; upward because charge is positive.

Video Solution:

Q5 **Text Solution:**

Electric lines of force do not form closed loops.

Video Solution:

Q6 **Text Solution:**

Electric lines of force do not intersect each other, as this would imply that the electric field has two directions at a point, which is impossible.

Video Solution:

Q7 **Text Solution:**

The electric field lines due to a positive point charge radiate outward from the charge.

Video Solution:

Q8 **Text Solution:**

Lines are denser at A. so $\mathbf{E_A} > \mathbf{E_B}$

Video Solution:

Q9 **Text Solution:**

Both are negative

Video Solution:

Q10 **Text Solution:**

The assertion is false because the electric field lines between two point charges are not always straight, especially when the charges are of unequal magnitudes or the distance between them is large. They are straight only in the case of equal magnitude charges at a large distance apart. The reason, however, is true: electric field lines always point from positive to negative charges.

Video Solution:

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